Task Time Detective

How long does each turnaround task take, and when is the car ready?

We estimate all 9 task durations for every car, and the total time until it is ready,
by rebuilding how the branch actually works: a staff schedule per car, a first-come-first-served queue between cars, and opening hours.

Total turnaround MAE

7.7 min

baseline 99 min · 13× smaller

Per-task WAPE

0.075

baseline ~0.55 (README) · 'Strong' < 0.2

Mean per-task MAE

2.67 min

baseline 17 min · all 9 tasks equal weight

Validation week, leave-one-day-out (no model saw the day it predicts).

Held-out train weeks: total MAE 5.8. Expected on the released test weeks: ~5.5 min.

0 negative or impossible durations (the baseline gives 1,575 negative ones on validation).

How a turnaround is built

Found by checking the true task times in the labelled week

Car returned
return time,
car condition, tasks

→

Key box
branch closed?
work starts at opening

→

Queue
first come, first served;
≈ 12 min ÷ staff
per car ahead

→

Work on the car
fixed priority schedule
(below)

→

Ready
work pauses at closing,
resumes at next opening

The schedule inside each car (stations and airports)

1. A damage repair comes first: damage check, then repair, car alone.
2. Other tasks go out in a fixed order to whichever worker is free first:
   exterior → special → tyre → interior → damage check → refuel.
3. Interior cleaning waits for special cleaning (both inside the car).
4. Electric charging runs alongside on the charger.

Evidence

97–98%

of quiet-branch totals reproduced
exactly (82% for longest-task-first)

Fixing it cut station errors
from 8.1 to 5.5 min.

Per-task estimates: effects simply add up

Model: scikit-learn QuantileRegressor (median linear regression, L1 penalty), one per task

Tyre check: median per staffing level · Damage repair: HistGradientBoostingRegressor, also learned from ~1,100 training totals

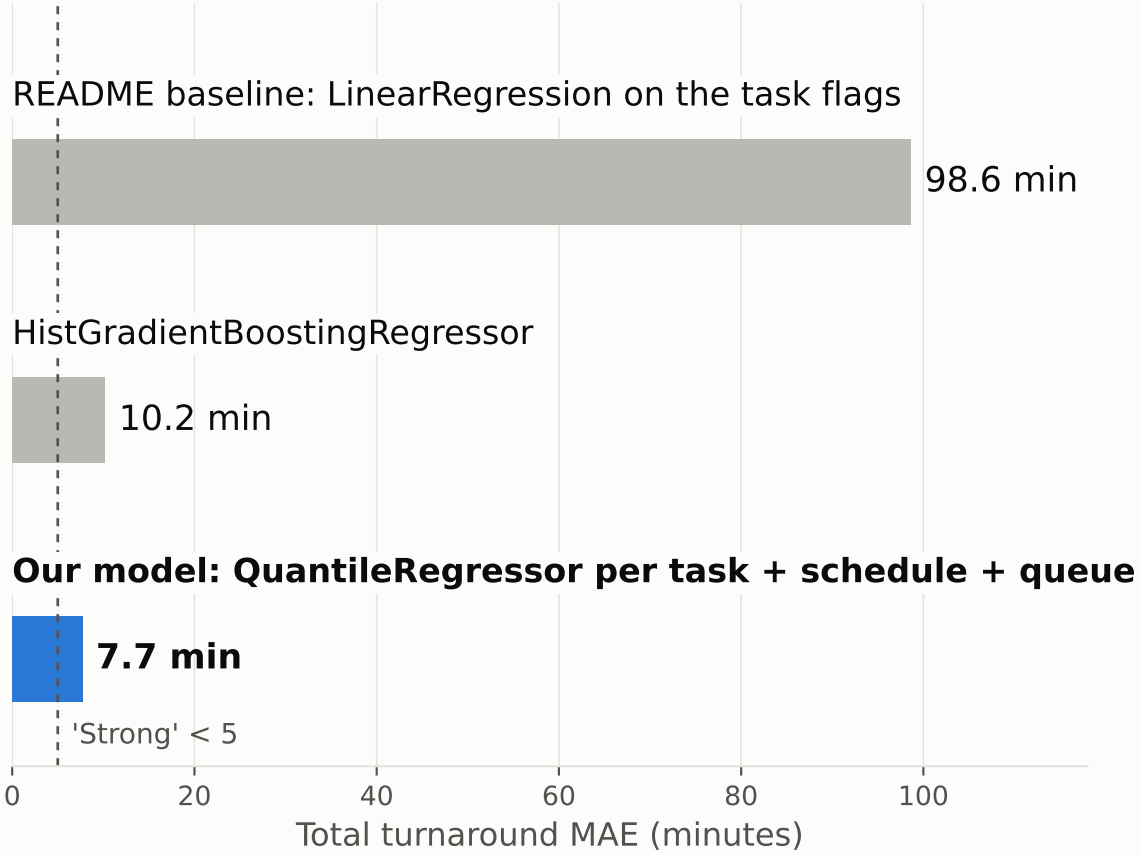
Task	Avg (min)	What drives it (minutes added)
Interior cleaning	16.4	child seat +7.6; dirt 2/3/4/5: +2.2 / +4.4 / +6.6 / +8.5; van +1.8
Normal exterior	19.2	dirt 2/3/4/5: +1.8 / +3.6 / +5.4 / +7.4; van +2.5; rain +1.6
Quick exterior	13.3	dirt 3/4/5: +0.9 / +1.3 / +2.1; airports +0.9 to +1.2
Special cleaning	30.0	smoking +7.9; premium +2.7; pet +2.0; child seat +1.6
Charging	24.8	only battery level: ≈ 40 − 0.34 min per % charged
Refuel	13.5	branch (pump distance): FRA +6.1 … NUE −3.8; van +4.1, SUV +2.9
Damage check	2.4	only reported damage: +0.3
Damage repair	68.6	reported damage ≈ +20; the rest is noise
Tyre check	26.1	47 / 33 / 17.5 min at 1 / 2 / 3-staff branches

A child seat adds the same ~7.6 min at every dirt level, so a 45-coefficient linear model beats boosted trees on all 7 frequent tasks.

Results vs the baseline

Validation week, leave-one-day-out · same rows and metrics for every model

Total time until the car is ready

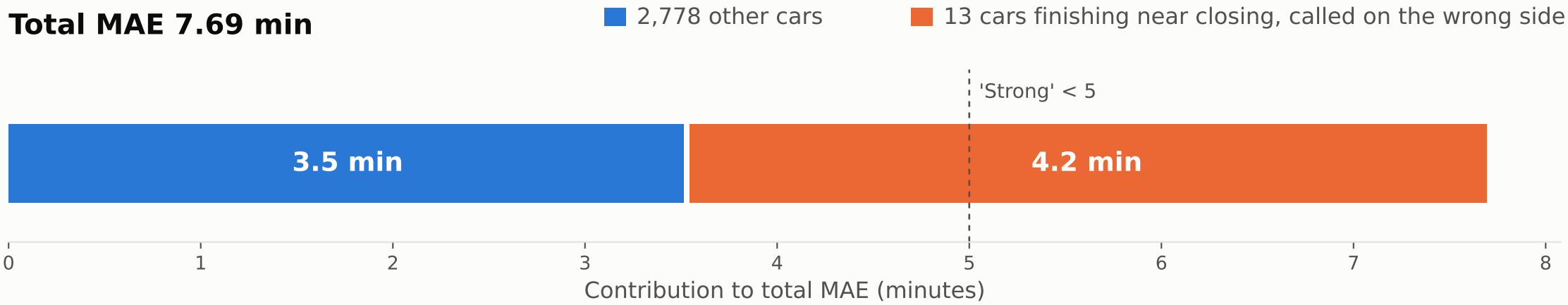


Per-task MAE (minutes)

Task	Baseline	Ours
Interior cleaning	16.4	**1.24**
Normal exterior	16.1	**1.61**
Quick exterior	16.4	**0.80**
Special cleaning	22.9	**0.80**
Charging	24.1	**1.63**
Refuel	7.3	**0.86**
Damage check	1.8	**0.20**
Damage repair	32.1	**12.62**
Tyre check	16.5	**4.25**
Mean of 9 tasks	17.1	**2.67**
WAPE	~0.55	**0.075**

Why the total stops around 6–8 minutes

Where the remaining total error comes from (validation week)



Queue stays are random. Cars are served first come, first served, but how long each one stays in the queue is not predicted by anything in the data (best departure rule: 53% exact; car features make it worse).

Closing time amplifies it. For ~2% of cars a few random minutes decide 'ready tonight' vs 'tomorrow morning': a wrong call costs 6 hours (airport night) up to a whole weekend (city centers).

The model is at its floor. Its own predicted uncertainty implies 5.9 min on held-out train weeks; it achieves 5.8. Expected on the released test weeks: ~5.5 min (range 5.4–7.2).